

Jason Ives

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Academic Portfolio: <https://jasonives.github.io/Portfolio/> | LinkedIn: www.linkedin.com/in/jason-r-ives

Professional Summary

Senior Data Systems Architect & Master of Data Science specializing in predictive modeling, deep learning architectures, and scalable data pipeline engineering. Over two decades of experience designing high-stakes automation, modernizing data architectures, and leading technical initiatives across international operations. Combining deep system expertise into applied AI research, LLM ensemble optimization, performant Rust systems programming, and production machine learning solutions.

Education

University of Guelph | Guelph, ON — Master of Data Science | Geospatial Stream

Arizona State University | Tempe, AZ — B.S. in Psychology | Minor in Philosophy

Core Competencies

- **Machine Learning & AI:** Scikit-learn, PyTorch, Deep Learning, LLM Mechanics & Multi-Agent Ensembles, Predictive Analytics, Model Evaluation & Performance Tuning.
- **Data Engineering & Languages:** Python (Advanced), Rust, SQL, R, PySpark, ETL Pipeline Optimization, Git Workflows, System Modernization.
- **Quantitative & Analytical Methods:** Statistical Modeling, Spatiotemporal Simulation, Network Optimization, High-Dimensional Data Processing.
- **Leadership & Infrastructure:** Data Governance, Technical Operations, Software Lifecycle Design, Edge-to-Cloud System Architecture.

Technical & Applied AI Projects

V-BReE: Built a multi-agent refinement framework using Python, focusing on iterative response optimization and reduced intra-ensemble sycophancy to generate high-accuracy model responses. Highlights include achieving 6.8% correctness on items where 0/3 constituent models achieved correct responses, and 61% recovery of correct responses on items where 1/3 constituent models achieved correct responses.

LiDAR Point Cloud Deep Learning System: Developed a high-performance, lightweight deep learning model in Rust capable of classifying millions of LiDAR points with 84.7% accuracy in minutes. Optimized the system for low computational overhead and open-source integration, utilizing AI-assisted development workflows (Cline, Claude) to accelerate code generation and performance tuning.

Ground Logistics Dispatch Engine: Designed and built a weather-centric routing dashboard that dynamically optimizes truck dispatching by translating real-time meteorological data streams into road-impedance cost networks.

ST-MELD: Developed a predictive simulation model using Python to analyze complex fluid flow behavior, demonstrating advanced capabilities in physical-process modeling and numerical simulation.

Professional Experience

Senior CPro Programmer | Columbia University ICAP | New York, NY | 2023 – 2026

- Lead the programming and system architecture of tablet-based edge-to-cloud data instruments supporting population health studies across 25,000+ households.
- Architected offline-first data pipelines to ensure zero data loss in severe low-connectivity and logistically constrained environments.
- Conducted rigorous code testing, automated QA validation, and developer oversight to enforce strict data integrity standards across international technical teams.

Senior Research Operations Associate | Westat | Rockville, MD | 2004 – 2023

- Designed and built Python-based tooling to automate complex data processing pipelines, improving localization efficiency by over 50%.
- Established internal development divisions and provided technical oversight for custom software tools and high-throughput data extraction systems.
- Modernized institutional software lifecycles by introducing Git-based version control and structured code review workflows.
- Researched, benchmarked, and executed enterprise platform migrations translating legacy database structures into modern cloud environments.
- Managed technical deployments and provided hands-on engineering guidance across field sites in Africa and South Asia.